

Project Progress Report: Enhancing Skeleton-Guided Native 3D Generation via Modality Compatible SK-Adapter

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CS-503 Project Progress Report 1

I. MILESTONE PROGRESS

For this milestone, our primary focus was on the exploring the capabilities of the SK-Adapter and establishing a robust workflow for skeleton-guided 3D generation. We have successfully trained the SK-Adapter model.

We have explored multiple conditional generation settings and successfully synthesized preliminary results. As demonstrated below, our model can effectively perform image generation guided by structural constraints, specifically:

- **Skeleton + Text:** Generating images conditioned on structural skeletons and descriptive text prompts (see Fig. 1).
- **Skeleton + Image:** Generating images conditioned on skeleton structures and reference images (see Fig. 2).

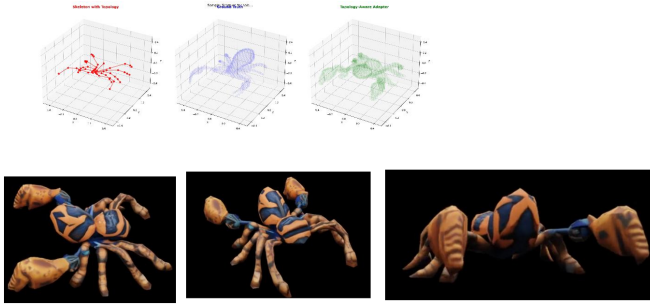


Fig. 1. Skeleton conditioned 3D generation based on text: (SK-Adapter-text).

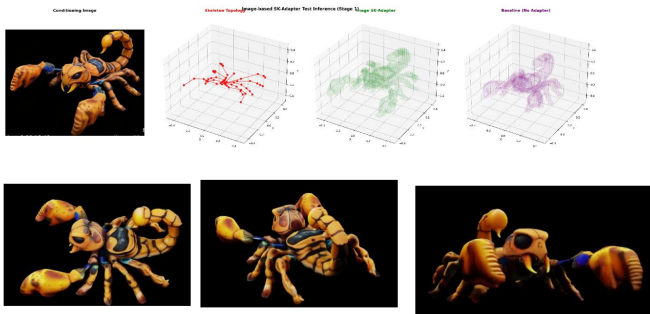


Fig. 2. Skeleton conditioned 3D generation based on image: (SK-Adapter image).

Furthermore, to enhance the visual quality and semantic alignment of the generated images, we have experimented with incorporating an external Large Language Model (LLM) combined with image editing techniques (using the Seedream 2.0 image editing model) to further optimize and refine the final outputs, as shown in Fig. 3.

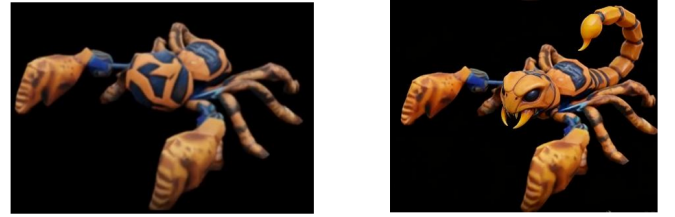


Fig. 3. Image enhancement and optimization utilizing an external LLM and the Seedream 2.0 image editing model.

These concrete steps prove the feasibility of our proposed approach.

II. DISCUSSION & NEXT STEPS

Our progress so far aligns well with our project goals regarding model training and concept validation. The generated images utilizing the SK-Adapter demonstrate fine-grained structural control. However, we also identified several challenges and missing components during this phase.

Currently, the complete generation workflow has not been fully integrated into a cohesive pipeline. The processes of utilizing the SK-Adapter, querying the external LLM, and applying image editing are still relatively disjointed. Furthermore, we have not yet conducted comprehensive evaluations of the generated results.

To address these gaps, our focuses for the next phase are as follows:

- **Workflow Integration:** Build a generation framework that connects the SK-Adapter, the external LLM agent, and the image editing modules without manual intervention.
- **Benchmarking:** Run systematic evaluations on Objaverse-TMS to validate the robustness of our approach and compare it against baseline models. We evaluate on the following metrics: ReRigging Score, CLIP Score, KD-Dino and Pickscore.

III. AUTHOR CONTRIBUTION STATEMENT

- **Yuzhuo Ao:** Handled the training and debugging of the SK-Adapter model.
- **Yuanzhong Chen:** Explored the integration of the external LLM and image editing optimizations.
- **Tingsen Liu:** Prepared the demonstration results and will lead the workflow construction.
- **Zitian Wang:** Evaluation framework setup and benchmarking.